



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Multi AI Agents Workflow for Content Creation

TGS Ref No: TGS-2023036153

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 2.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	13 Sep 2026	Multi-agent content strategy courseware aligned to the current course page and TSC; GreenLeaf Café case; 12 labs.	Tertiary Infotech Academy
2.0	8 October 2026	Revamped in the house design of Agentic AI Applications with Codex (TGS-2023041081). One central case — Horizon Wealth Planning, a Singapore financial-planning firm — across 14 labs, each in its own folder: Claude Cowork sub-agents (research, ideation, personas, cadence, skills, measurement), Codex (website, compliance subagent, approval gate, video), Claude Code subagents and agent teams (social, newsletter and video teams with LinkedIn, Facebook and YouTube publishing behind a human approval gate) and Hermes Agent bots on a Kanban board. Practice Exam slide (Claude Certified Associate — Foundations). WA (6 SAQ, K1-K6) and PP (4 tasks, A1-A7) rewritten for the v2.0 labs; assessment 4:00-6:00 PM Day 2 (WA 1 h, PP 1 h).	Dr. Alfred Ang

TABLE OF CONTENTS

Course Overview	4
Learning Outcomes	4
Daily Schedule	4
Day 1 — Research, Build and Plan.....	4
Day 2 — Create, Distribute and Govern.....	5
Topic-by-topic Breakdown	6
Topic 1 — Multi-AI-Agent Content Ideation and Digital Storyboarding · Slides 18–53.....	6
Lab 1 — A Research Team of Sub-agents · Slides 27–31.....	6
Lab 2 — Plan and Build the Horizon Website with Codex · Slides 35–38.....	7
Lab 3 — A Compliance Subagent, Then Publish · Slides 41–45.....	7
Lab 4 — Ideation Sub-agents and a Digital Storyboard · Slides 48–52.....	7
Topic 2 — Audience Research and Content Requirement Analysis · Slides 54–77.....	7
Lab 5 — Audience Research and Personas from Evidence · Slides 58–62.....	7
Lab 6 — Content Cadence and the Editorial Calendar · Slides 64–68.....	8
Lab 7 — Studio Skills: Voice, Facts, Compliance and Formats · Slides 72–76.....	8
Topic 3 — Multi-Channel Content Creation and Agent Workflow Coordination · Slides 78–111.....	8
Lab 8 — The Human Approval Gate and the Publishers · Slides 86–90.....	8
Lab 9 — The Social Media Agent Team: LinkedIn and Facebook · Slides 92–97.....	8
Lab 10 — The Newsletter Agent Team: the Sunny Sunday Email · Slides 99–103.....	9
Lab 11 — The Video Agent Team: Script to YouTube · Slides 105–110.....	9
Topic 4 — Content Distribution, Strategy Guidelines and Responsible AI Practices · Slides 112–142.....	9
Lab 12 — Hermes Agent Bots on a Kanban Board · Slides 117–121.....	9
Lab 13 — Always On: a Telegram Bot That Asks Before It Acts · Slides 123–126.....	10
Lab 14 — Measure, Govern and Write the Playbook · Slides 129–133.....	10
Resources Required	10
Assessment	11

Course Overview

A two-day, hands-on WSQ course in content strategy with multi-agent AI workflows. Learners run one business throughout — Horizon Wealth Planning, an independent financial-planning firm in Singapore — and build its content studio of agents: Claude Cowork sub-agents for research, ideation, audience analysis, cadence and measurement; Codex for the website, a compliance subagent, the human approval gate and the video; Claude Code subagents and agent teams for the social, newsletter and video teams; and Hermes Agent bots on a Kanban board for always-on delivery. Day 1 researches, builds and plans; Day 2 creates, distributes and governs, with a person approving every external piece.

Course code TGS-2023036153 · 2 days · 16 hours including a 2-hour assessment · 142-slide deck v2.0 · 14 labs, each in its own folder. TSC: ICT-SNM-4004-1.1 Content Strategy.

Learning Outcomes

By the end of this course, participants will be able to:

- **LO1.** Conceptualize content ideas to meet marketing objectives and map out digital storyboards as part of a content strategy.
- **LO2.** Identify content requirements from customer preferences and determine the frequency of marketing content.
- **LO3.** Determine content types and styles and decide on modes and processes for distributing content.
- **LO4.** Develop guidelines for content strategy execution using appropriate delivery modes and responsible AI practices.

Daily Schedule

Both days run 9:00 AM – 6:00 PM with two 10-minute tea breaks and a 45-minute lunch. The Slides column gives the deck pages for each teaching block.

Day 1 — Research, Build and Plan

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:20 AM	20 min	Welcome, attendance, trainer, ground rules, learning outcomes, the Horizon scenario, lesson plan, assessment briefing	Presentation, discussion	1–17
9:20 AM–10:00 AM	40 min	Topic 1 — from prompts to agent teams; the four tools; Claude Cowork and sub-agents; the research team	Presentation, live demo	18–26
10:00 AM–10:45 AM	45 min	Lab 1 — A research team of sub-agents	Hands-on lab	27–31
10:45 AM–10:55 AM	10 min	Tea break	—	—
10:55 AM–11:10 AM	15 min	Codex: projects, /plan and AGENTS.md; the website brief	Presentation, live demo	32–34
11:10 AM–11:55	45 min	Lab 2 — Plan and build the Horizon	Hands-on lab	35–38

AM		website with Codex		
11:55 AM–12:05 PM	10 min	Financial content is regulated; a reviewer that only reports	Presentation	39–40
12:05 PM–12:40 PM	35 min	Lab 3 — A compliance subagent, then publish	Hands-on lab	41–45
12:40 PM–1:25 PM	45 min	Lunch	—	—
1:25 PM–1:40 PM	15 min	From objective to storyboard: diverge, score, decide	Presentation	46–47
1:40 PM–2:20 PM	40 min	Lab 4 — Ideation sub-agents and a digital storyboard	Hands-on lab	48–52
2:20 PM–2:30 PM	10 min	Topic 1 recap	Discussion	53
2:30 PM–2:45 PM	15 min	Topic 2 — evidence before personas; data agents count	Presentation	54–57
2:45 PM–3:25 PM	40 min	Lab 5 — Audience research and personas from evidence	Hands-on lab	58–62
3:25 PM–3:35 PM	10 min	Tea break	—	—
3:35 PM–3:45 PM	10 min	Finding the right frequency; capacity and review buffers	Presentation	63
3:45 PM–4:20 PM	35 min	Lab 6 — Content cadence and the editorial calendar	Hands-on lab	64–68
4:20 PM–4:35 PM	15 min	Agent skills: one skill set for every tool	Presentation, live demo	69–71
4:35 PM–5:15 PM	40 min	Lab 7 — Studio skills: voice, facts, compliance and formats	Hands-on lab	72–76
5:15 PM–6:00 PM	45 min	Day 1 review, Q&A and catch-up	Discussion	77

Day 2 — Create, Distribute and Govern

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:20 AM	20 min	Topic 3 — subagents vs agent teams; human in the loop, enforced; the publishing APIs	Presentation, live demo	78–85
9:20 AM–9:55 AM	35 min	Lab 8 — The human approval gate and the publishers	Hands-on lab	86–90
9:55 AM–10:00 AM	5 min	The social media team	Presentation	91
10:00 AM–10:45 AM	45 min	Lab 9 — The social media agent team: LinkedIn and Facebook	Hands-on lab	92–97
10:45 AM–10:55 AM	10 min	Tea break	—	—
10:55 AM–11:00 AM	5 min	The newsletter team; consent and email rules	Presentation	98
11:00 AM–11:40 AM	40 min	Lab 10 — The newsletter agent team: the Sunny Sunday email	Hands-on lab	99–103
11:40 AM–11:45 AM	5 min	The video team; making and disclosing the video	Presentation	104

11:45 AM–12:25 PM	40 min	Lab 11 — The video agent team: script to YouTube	Hands-on lab	105–110
12:25 PM–12:30 PM	5 min	Topic 3 recap	Discussion	111
12:30 PM–1:15 PM	45 min	Lunch	—	—
1:15 PM–1:30 PM	15 min	Topic 4 — Hermes Agent, profiles and the Kanban board	Presentation, live demo	112–116
1:30 PM–2:15 PM	45 min	Lab 12 — Hermes agent bots on a Kanban board	Hands-on lab	117–121
2:15 PM–2:20 PM	5 min	Always-on delivery: gateway bots and schedules	Presentation	122
2:20 PM–2:50 PM	30 min	Lab 13 — Always on: a Telegram bot that asks before it acts	Hands-on lab	123–126
2:50 PM–3:00 PM	10 min	Tea break	—	—
3:00 PM–3:10 PM	10 min	Responsible AI for financial content; measuring what matters	Presentation	127–128
3:10 PM–3:50 PM	40 min	Lab 14 — Measure, govern and write the playbook	Hands-on lab	129–133
3:50 PM–4:00 PM	10 min	Course summary: why delegate to agents, how each pattern is triggered; Q&A and assessment briefing	Discussion	134–135
4:00 PM–5:00 PM	60 min	Written Assessment (WA) — open book, 1 hour	Assessment	—
5:00 PM–6:00 PM	60 min	Practical Performance (PP) — 1 hour; TRAQOM, attendance, submit, sign the record	Assessment	136–142

Topic-by-topic Breakdown

Topic 1 — Multi-AI-Agent Content Ideation and Digital Storyboarding • Slides 18–53

- From one chatbot to teams of agents
- Claude Cowork, Codex, Claude Code and Hermes — and the agent loop
- Research with sub-agents; build and check the website
- Ideas from parallel agents, then a five-beat storyboard

Lab 1 — A Research Team of Sub-agents • Slides 27–31

- **Surface:** Claude Cowork
- **Time:** 45 minutes (Day 1)
- **Features practised:** sub-agents in parallel · market research · business analysis · competitive analysis · a recommendation
- **Output:** research/: market research, business analysis, competitive analysis and a recommendation, every figure cited
- **Evidence:** Cowork confirmed all four roles before starting. No uncited figure; no INTERNAL fee anywhere in the reports.

- **Lab folder:** labs/lab-01-research-team-of-subagents/

Lab 2 — Plan and Build the Horizon Website with Codex • Slides 35–38

- **Surface:** Codex
- **Time:** 45 minutes (Day 1)
- **Features practised:** Add a folder · /plan · AGENTS.md (/init) · build from data, not memory
- **Output:** docs/index.html: hero, services, retirement calculator, checklist, reviews, FAQ, contact form with PDPA consent
- **Evidence:** Plan mode showed steps AND questions before any edit. No horizontal scroll at 375px; AGENTS.md is under 60 lines.
- **Lab folder:** labs/lab-02-build-the-website-with-codex/

Lab 3 — A Compliance Subagent, Then Publish • Slides 41–45

- **Surface:** Codex → GitHub Pages
- **Time:** 35 minutes (Day 1)
- **Features practised:** a reviewer subagent that only reports · the compliance checklist · GitHub Pages from /docs
- **Output:** review/site-compliance.md (FAIL, then PASS) and the site live on GitHub Pages
- **Evidence:** The reviewer ran as a subagent and edited nothing. The Pages URL works on your phone.
- **Lab folder:** labs/lab-03-compliance-subagent-and-publish/

Lab 4 — Ideation Sub-agents and a Digital Storyboard • Slides 48–52

- **Surface:** Claude Cowork (sub-agents, Live Artefact)
- **Time:** 40 minutes (Day 1)
- **Features practised:** three ideators in parallel · rubric scoring · a human concept checkpoint · a five-beat storyboard
- **Output:** strategy/ideas.md (15 scored ideas), strategy/storyboard.md and an interactive storyboard Live Artefact
- **Evidence:** Three ideator sub-agents ran in parallel. The storyboard Live Artefact is saved in your sidebar.
- **Lab folder:** labs/lab-04-ideation-and-storyboard/

Topic 2 — Audience Research and Content Requirement Analysis • Slides 54–77

- Personas from evidence, not instinct
- Content requirements every agent can follow
- How often to publish: value, fatigue and capacity
- Skills: the studio's know-how, written once

Lab 5 — Audience Research and Personas from Evidence • Slides 58–62

- **Surface:** Claude Cowork (sub-agents)
- **Time:** 40 minutes (Day 1)
- **Features practised:** one analyst sub-agent per data source · counts, not impressions · confidence labels · content requirements
- **Output:** strategy/personas.md (three personas with evidence) and strategy/content-spec.md

- **Evidence:** Three analyst sub-agents ran, one per file. content-spec.md has every field for every persona.
- **Lab folder:** labs/lab-05-audience-research-and-personas/

Lab 6 — Content Cadence and the Editorial Calendar • Slides 64–68

- **Surface:** Claude Cowork
- **Time:** 35 minutes (Day 1)
- **Features practised:** marginal value vs fatigue · team capacity · review buffers · a four-week calendar
- **Output:** strategy/cadence.md and strategy/calendar.csv (four weeks from 2 November 2026)
- **Evidence:** cadence.md gives a frequency per channel with the evidence. No week is over the team's 12 hours.
- **Lab folder:** labs/lab-06-cadence-and-editorial-calendar/

Lab 7 — Studio Skills: Voice, Facts, Compliance and Formats • Slides 72–76

- **Surface:** Claude Cowork (/skill-creator) → every tool
- **Time:** 40 minutes (Day 1)
- **Features practised:** /skill-creator · SKILL.md · one skill set shared by Cowork, Claude Code, Codex and Hermes
- **Output:** Four skills — sunny-voice, fact-check, fin-compliance, channel-formats — in skills/, .claude/skills/ and .agents/skills/
- **Evidence:** Four skills appear under Customize → Skills in Cowork. fact-check marked a figure not on the facts sheet UNVERIFIED.
- **Lab folder:** labs/lab-07-studio-skills/

Topic 3 — Multi-Channel Content Creation and Agent Workflow Coordination • Slides 78–111

- Subagents or an agent team?
- Human in the loop — enforced, not requested
- Social, newsletter and video teams
- From approval to LinkedIn, Facebook, email and YouTube

Lab 8 — The Human Approval Gate and the Publishers • Slides 86–90

- **Surface:** Codex
- **Time:** 35 minutes (Day 2)
- **Features practised:** approve.mjs (people only) · publish.mjs (dry run by default) · a hook that blocks agents · LinkedIn, Facebook and YouTube APIs
- **Output:** scripts/submit.mjs, approve.mjs, publish.mjs, gate-hook.mjs, the hook in .claude/settings.json, and review/approvals.csv
- **Evidence:** Codex could not approve: approve.mjs refused without a terminal. .env is git-ignored and holds no token you pasted into a chat.
- **Lab folder:** labs/lab-08-approval-gate-and-publishers/

Lab 9 — The Social Media Agent Team: LinkedIn and Facebook • Slides 92–97

- **Surface:** Claude Code (subagents + agent teams)

- **Time:** 45 minutes (Day 2)
- **Features practised:** subagent definitions in `.claude/agents/` · an agent team with a shared task list · teammates message each other · human review → auto-post
- **Output:** `content/social/week-01/`: 2 LinkedIn and 3 Facebook posts, reviewed, approved by you and posted (or dry-run)
- **Evidence:** Three subagent files exist, each with tools and a model. Dry runs printed every request; live posts appear only if you said go.
- **Lab folder:** `labs/lab-09-social-media-agent-team/`

Lab 10 — The Newsletter Agent Team: the Sunny Sunday Email · Slides 99–103

- **Surface:** Claude Code (agent team) → Claude Cowork (Gmail)
- **Time:** 40 minutes (Day 2)
- **Features practised:** an agent team with a task that waits for a person · consent filtering · HTML and text email · Gmail drafts
- **Output:** `content/newsletter/2026-11/`: `issue.md`, `newsletter.html`, `newsletter.txt`, `recipients.csv`, `excluded.csv`, and Gmail drafts
- **Evidence:** Four teammates; the builder waited for your approval. Your test email reads well on a phone.
- **Lab folder:** `labs/lab-10-newsletter-agent-team/`

Lab 11 — The Video Agent Team: Script to YouTube · Slides 105–110

- **Surface:** Claude Code (agent team) → Codex (render, upload)
- **Time:** 40 minutes (Day 2)
- **Features practised:** agent team for the script · approval of script and video · ffmpeg render · private YouTube upload
- **Output:** `content/video/ep01/`: `scenes.json`, `ep01.mp4`, `ep01.md`, and a private YouTube upload (or dry run)
- **Evidence:** The reviewer passed the script; you approved `yt-ep01-script`. The upload is private (or a dry run printed the request).
- **Lab folder:** `labs/lab-11-video-agent-team-to-youtube/`

Topic 4 — Content Distribution, Strategy Guidelines and Responsible AI Practices · Slides 112–142

- Hermes Agent: open-source agent bots on any model
- A Kanban board for a team of agents
- Always-on delivery: schedules and chat bots
- Measure, govern and write the playbook

Lab 12 — Hermes Agent Bots on a Kanban Board · Slides 117–121

- **Surface:** Hermes Agent (profiles, Kanban)
- **Time:** 45 minutes (Day 2)
- **Features practised:** open-source Hermes Agent · one profile per role · a Kanban role pipeline · review and human unblock · the same skills
- **Output:** Three Hermes profiles, a researcher → writer → reviewer pipeline on the board, and Week 2 content approved by you

- **Evidence:** hermes profile list shows researcher, writer and reviewer. You unblocked it; the post is submitted to approvals.csv.
- **Lab folder:** labs/lab-12-hermes-agent-bots-on-kanban/

Lab 13 — Always On: a Telegram Bot That Asks Before It Acts • Slides 123–126

- **Surface:** Hermes Agent (gateway bot, cron)
- **Time:** 30 minutes (Day 2)
- **Features practised:** a Telegram bot · scheduled jobs in plain words · delivery targets · approve from your phone · pause
- **Output:** A Telegram bot, a Monday 8am content-brief job, and one approved brief from your phone
- **Evidence:** The bot answers only you. The job is paused after class.
- **Lab folder:** labs/lab-13-always-on-telegram-bot/

Lab 14 — Measure, Govern and Write the Playbook • Slides 129–133

- **Surface:** Claude Cowork (sub-agents, Live Artefact)
- **Time:** 40 minutes (Day 2)
- **Features practised:** analyst sub-agents · cost per chat · comparing agent patterns · responsible-AI controls · the execution playbook
- **Output:** A results dashboard (Live Artefact), reports/agent-comparison.md and strategy/content-strategy-playbook.md
- **Evidence:** The dashboard shows cost per chat for every channel. It has an incident procedure, and your own rule is in it.
- **Lab folder:** labs/lab-14-measure-govern-playbook/

Resources Required

- Trainer: the v2.0 slide deck, this Lesson Plan, the Learner Guide and the lab pack (labs/, 14 folders, with reference solutions for Labs 2, 3, 7, 8, 9 and 11).
- Each learner: a laptop with the Claude desktop app on a Pro plan or above (Claude Cowork and Claude Code), the ChatGPT desktop app on a plan that includes Codex, Node.js 22, Python 3 with Pillow, ffmpeg, Git, a GitHub account and a personal Google account (Gmail).
- A spend-limited Anthropic or OpenRouter API key issued by the trainer for Hermes Agent (Labs 12-13), revoked after class.
- Optional, for live publishing: a LinkedIn developer app, a test Facebook Page and a Google Cloud project with the YouTube Data API. Every lab also works in dry-run mode without them.
- Optional: a Telegram account for the Hermes bot (Lab 13).
- Internet access to claude.ai, chatgpt.com, github.com, api.linkedin.com, graph.facebook.com, googleapis.com and hermes-agent.nousresearch.com.
- All fixtures are synthetic. Every learner email address resolves to the learner's own Gmail via plus-addressing.

Assessment

- Day 2, 4:00 – 6:00 PM: Written Assessment (WA, short-answer, 1 hour) from 4:00 PM, then Practical Performance (PP, 1 hour) from 5:00 PM.
- Open book: slides, Learner Guide and approved materials only.
- Funding criteria: minimum 75% attendance, and assessed as Competent in both the WA and the PP.
- Assessment flow: TRAQOM survey → assessment digital attendance → WA and PP → submit answers on the LMS → sign the Assessment Summary Record.
- Courseware and assessment are on the LMS: <https://lms-tms.tertiaryinfotech.com/>